



Nexcomm Systems · Wi-Fi HaLow

Convenience Store / Gas Station

Nexcomm HaLow + mooreVIEW — cold chain, RTU, lift, leaks

One HaLow AP covers the shop and forecourt — planned WOs before spoilage or overflow

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At a Glance

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Convenience Store / Gas Station — Nexcomm

HaLow Story

Vertical: Convenience store / cold chain · Marketplace #5 (refrigeration overlap)
Nexcomm reference: Gas station / convenience store slide (shop + forecourt)
MV project: cstore-halow (fixture scaffold)
Pricing: Commercial tier — \$2,500 + \$25/mo (+ \$3/mo cell) · Year-1 \$2,836

One-line pitch

One Nexcomm HaLow AP covers the **café, walk-in, bathrooms, grease/lift, and forecourt** — miles better than 2.4 GHz for a gas-station footprint. mooreVIEW turns cooler drift, RTU faults, and lift level into **planned work orders before product loss or overflow**.

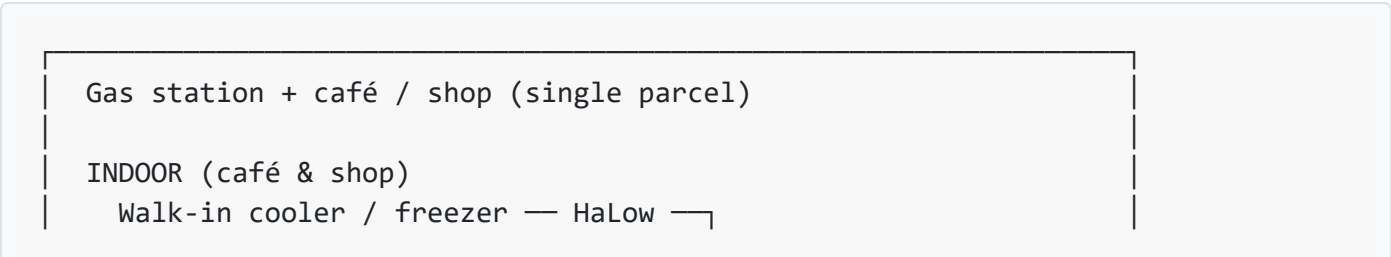
Reactive vs proactive

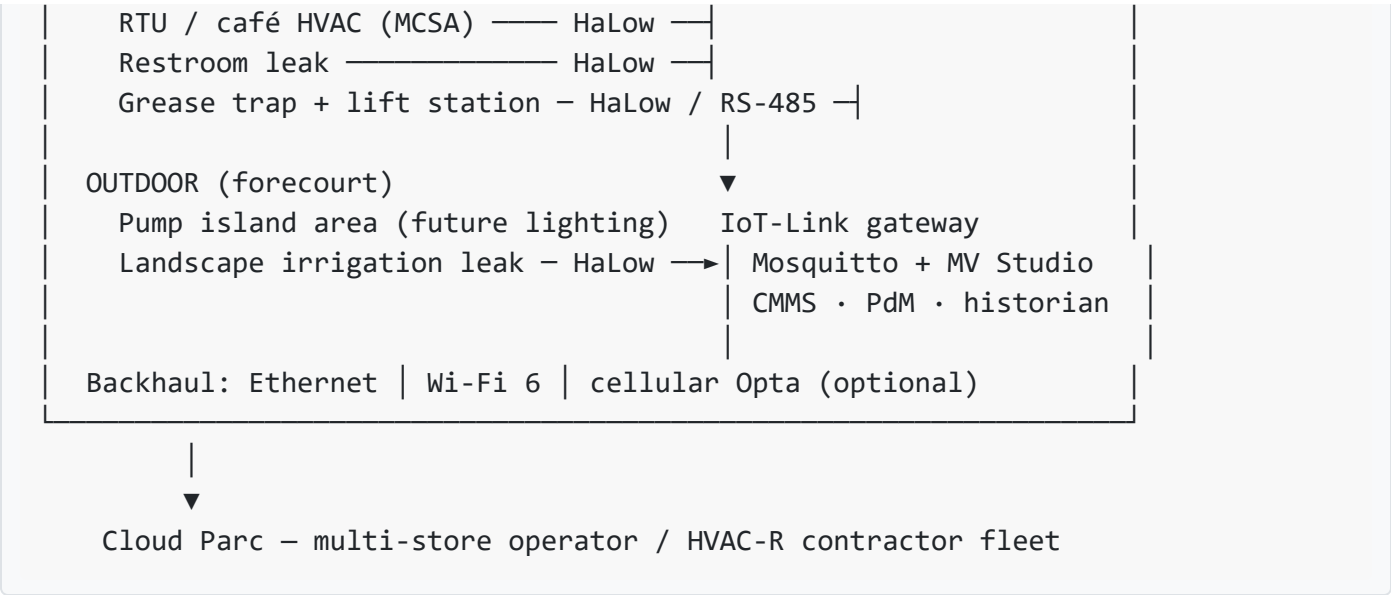
Reactive (today)	Proactive (mooreVIEW + Nexcomm)
Warm box found during restock	Walk-in/rack trend → WO before spoilage
RTU locked out — no cool in café	MCSA compressor/fan advisory → tune before fail
Grease overflow / lift high-level call	Level + pump runtime → WO before SSO
Bathroom leak after ceiling damage	HaLow leak tag → critical WO same day
Siloed OEM refrigeration app	One .est + contractor Parc across the chain

Site architecture

Nexcomm deck pattern: **HaLow sensors site-wide** → **Nexus AP** → **optional cellular** → **cloud** → **laptop/tablet dashboards**.

mooreVIEW mapping:





Gateway: Compulab **IoT-Link** (or Nexcomm **Nexus AP** with MQTT bridge to MV). Single broker on-site; HaLow AP routes sensor LAN to broker IP.

For **duplex lift / grease** at the shop: reuse lift-station I/O (level, pump runtimes) on the same appliance or a small Opta panel on RS-485.

Node inventory (reference site)

Maps Nexcomm **gas station / c-store** diagram to MV + Nexcomm hardware.

Zone	Device ID	Platform	Driver / template	Monitors
Walk-in cooler	cstore_walkin_01	Nexcomm Halo or T-HaLow	nexcomm_halo_mqtt / Parc	Box temp, optional door
Reach-in freezer (café)	cstore_freezer_01	Nexcomm Halo	nexcomm_halo_mqtt	Freezer temp band
Beverage rack	cstore_rack_01	Modbus probe or Halo X	Modbus / Halo expansion	Rack temp ×N
Café RTU / split	cstore_rtu_mcsa	Nexcomm MCXN947	mcxn947_hvac_mcsa	COMP/FAN MCSA, T1–T4, pan leak
Restroom	cstore_rr_leak	Nexcomm HaLow leak	nexcomm_halow_leak_mqtt	CH1 leak, optional flow
Grease / lift	cstore_lift_01	Opta or IoT-Link I/O	mqtt_parc / Modbus	Level, pump1/2 run, starts
Forecourt irrigation	cstore_irrig_leak	Nexcomm HaLow leak	nexcomm_halow_leak_mqtt	CH2 landscape leak

Site gateway	mv_cstore_01	IoT-Link	—	Broker, ST, CMMS, uplink
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Fixture: `st/fixtures/cstore_halow.json` · CSV:
`st/fixtures/cstore_halow_nodes.csv`

Nexcomm slide → MV pack

Nexcomm application	MV implementation	Priority
Cooler / freezer monitoring	Cold Chain tags + bands	P0
HVAC monitoring	MCXN947 MCSA on RTU	P0
Sink / toilet leak	<code>nexcomm_halow_leak_mqtt</code>	P0
Grease trap & lift station	Lift-station .est slice on same gateway	P0
Outdoor lighting	Not in v1 — custom DO / phase 2	P2
Irrigation leak	HaLow leak channel on exterior node	P1

Proactive signal model

Leading indicator	Tag / logic	CMMS outcome
Walk-in temp drifting high	<code>WALKIN_TEMP</code> band persistence	Planned refrigeration WO
Compressor short-cycle count	ST counter on rack suction	WO before lockout
RTU fan MCSA fault	<code>RTU_FAN_FLT</code>	Tune/replace before café outage
RTU compressor MCSA fault	<code>RTU_COMP_FLT</code>	PdM-sourced PM
Lift lag runtime imbalance	<code>PUMP2_RUNTIME</code> VS <code>PUMP1</code>	WO before SSO
High level band	<code>WET_WELL_LEVEL</code>	Critical WO
Restroom leak	<code>RR_LEAK</code>	Critical WO — water damage
Door-open persistence (optional)	Door DI + temp delay	Nuisance-filtered high temp

ST advisories: door-aware high-temp delay; short-cycle counters; lift alternating runtime checks — same patterns as [refrigeration detailed plan](#) and lift-station pack.

Device templates

Template	Topic / transport	C-store use
nexcomm_halow_leak_mqtt	nexcomm/halow/<id>/telemetry	Restroom, irrigation, grease pan
nexcomm_halo_mqtt	nexcomm/halo/<id>/telemetry	Walk-in, freezer, rack expansions
nexcomm_bme688_env_mqtt	nexcomm/env/<id>/telemetry	Café air quality (optional)
mcxn947_hvac_mcsa	Parc mooreview/v1	Rooftop RTU / split condenser
Lift I/O (Opta / IoT-Link)	Parc or Modbus	Grease / duplex lift

Commissioning checklist

1. **Survey** — walk-in/rack count, RTU location, lift panel access, HaLow AP mount (cover shop + forecourt).
2. **Install IoT-Link** — `deploy/iot-link/` generic or c-store profile; enable MQTT Parc hub.
3. **HaLow AP** — gateway IP reachable from 915 MHz LAN.
4. **Cold chain nodes** — Halo or T-HaLow; set deviceId, broker, box bands in `.est`.
5. **RTU MCSA** — Nexcomm MCXN947; `MV_DEVICE_ID=cstore_rtu_mcsa` ; verify COMP_FLT/FAN_FLT in historian.
6. **Lift / grease** — import lift-station tags or RS-485 driver; tie high level to critical CMMS.
7. **Leak nodes** — map CH1 restroom, CH2 irrigation in `nexcomm_halow_leak_mqtt` driver.
8. **Cellular** (optional) — cellular Opta gateway or IoT-Link cell uplink; `$3/mo` line item.
9. **Alarm drill** — simulated high box temp + level band → WO with trend screenshot.
10. **Fleet** — clone `.est` to next store; Parc tenant for contractor route.

Acceptance: leading advisories create WOs; cold + lift online ≥99%; no nuisance high-temp during normal door use.

Multi-store contractor Parc

View	Filter
Food-critical	All stores with walk-in/rack alarm or drift
Lift / grease	High level or runtime imbalance
HVAC	RTU MCSA warning/critical
Today's route	Open proactive WOs sorted by geo

Clone `cstore-halow.est.json` per store; change `deviceId` prefix and site metadata only — portable `.est` model.

Proof metrics (12-month)

- ≥2 contractor or chain logos; ≥20 c-store sites on Parc.
 - Case study: **intervened before product loss** (walk-in drift WO, not emergency spoilage).
 - Lift: zero SSO events on monitored sites vs baseline.
 - Monitoring attach on existing PM / refrigeration contracts.
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Gaps vs Nexcomm deck

- **Outdoor lighting** monitor/control — not in MV marketplace v1; spec as Nexcomm DO module + future ST.
 - **Access control** — out of scope.
 - Wired **Modbus rack** vs wireless Halo — site survey chooses; MV supports both in one tag DB.
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Implementation status

Artifact	Status
This story	Done
<code>st/fixtures/cstore_halow.json</code>	Scaffold — node manifest
<code>scripts/cstore-halow/cstore-halow-data.js</code>	Scaffold — source for future generator
<code>cstore-halow.est.json</code>	Next — generate after cold-chain <code>.est</code> base exists
HMI screens	Reuse refrigeration + lift + RTU library widgets

Related files

Path	Purpose
<code>scripts/cstore-halow/cstore-halow-data.js</code>	Node inventory source
<code>st/fixtures/cstore_halow.json</code>	Site manifest
<code>src/devices/templates/nexcomm_*.json</code>	Nexcomm MQTT drivers

01-refrigeration-summary.md	Cold chain marketplace
04-lift-stations-summary.md	Grease / lift slice
MooreVIEW-Pricing-Guide.md	Commercial tier